

Credit Card Default Prediction Using Behavioral Time Series

1. Abstract:

This paper assesses whether tracking and analyzing credit card payments and consumer behavior spending over time can make an impact on payment protection insurance products. It also aims to determine whether early warning signals of financial trouble can be predicted, thereby allowing policyholders to intervene and undertake proactive measures before defaults occur. Using American Express credit card data covering nearly 450,000 U.S. customers and 13 months of transaction history from 2017 to 2018, this project develops models for predicting insurance underwriting and pricing that concentrate on behavioral trends rather than treating each month in isolation. To do so, the project models dynamic features that indicate whether cardholder behavior is improving or deteriorating, including trajectories of payment amounts, credit utilization growth rates, and payment timing consistency. It compares multiple modeling methods, including logistic regression with time-series features, random forests, and gradient-boosting models, and evaluates how they perform to detect patterns in consumer behavior that may represent increased likelihood of potential claim risk. The main aim of my project is to determine how far in advance defaults can be accurately predicted and to examine the trade-off between prediction lead time and accuracy of the model, which may directly impact the design of insurance intervention programs and risk reduction strategies. The main interest lies in identifying whether particular payment decline patterns emerge among future defaulters in the months leading up to actual default, such as decreasing payment amounts, increasing utilization, and greater payment irregularity, as these patterns would signal optimal intervention timing for insurance providers. Thus, integrating behavioral monitoring systems into insurance products can help both sides of parties by reducing insurance losses as well as providing some solutions to financially stressed customers through early intervention before defaulting.

2. Introduction:

Payment Protection Insurance (PPI) is a specialized coverage designed to assist borrowers in meeting their loan or credit card payments in the event of unexpected events, such as losing a job, becoming disabled, falling ill, or passing away (DataIntel, 2024). It serves as a financial safety net during economic downturns, helping to shield both consumers and lenders from the effects of default. Usually offered at the start of a loan, PPI can be purchased as a one-time premium added to the loan or paid in monthly installments alongside regular loan payments.

PPI accounts for an important and expanding share of the global insurance industry, valued at USD 22.4 billion in 2024 and projected to reach USD 34.7 billion by 2033, with a compound annual growth rate of 5.0% (DataIntel, 2024). Despite its large market share, current PPI products struggle with risk assessment methods. Traditional approaches rely on static risk models that assess creditworthiness only at the time of policy purchase, failing to capture the evolving behavioral signals indicating potential financial distress (DataIntel, 2024). This disconnect between fixed underwriting practices and the changing financial health of customers presents significant opportunities for insurers to reduce claim rates and improve customer outcomes through early detection. As consumer lending grows worldwide, particularly through credit cards and personal loans, the insurance industry faces increasing pressure to develop advanced, behavior-based risk assessment systems capable of identifying at-risk policyholders before default.

This project aims to address the gap by examining whether continuous monitoring of credit card payment and spending behavior can improve default prediction and facilitate early-warning systems for financial distress. Using American Express data from approximately 460,000 US customers over 13 months (March 2017-March 2018), we create predictive models that emphasize behavioral trends such as payment amount trajectories, credit utilization growth, and payment timing consistency rather than analyzing each month separately. We compare the effectiveness of logistic regression, random forest, and gradient-boosting approaches, assessing both prediction accuracy and practical lead time for insurance intervention strategies.

Recent studies illustrate both the opportunities and challenges of current methods. Abikoye et al. (2024) highlight how AI and ML are fundamentally transforming credit risk prediction, allowing financial institutions to go beyond traditional static models. Nayak (2024) supports incorporating behavioral analytics such as transaction patterns and payment habits into credit risk assessment for real-time evaluations and early warnings of financial distress. Similarly, Meriläinen (2023) shows that demographic factors and loan traits heavily influence PPI demand, endorsing behavioral segmentation for product development. However, there's limited research connecting these ML advancements to the practical design of insurance products, including pricing and intervention strategies.

Our analysis indicates that Gradient Boosting achieves an AUC of 94.8% in predicting default risk by leveraging behavioral time-series features. Notably, longitudinal studies of payment behavior reveal that defaulters show decreasing payment trends over the observation period (payments dropping by about 27% over 13 months) whereas non-defaulters tend to maintain or slightly increase their payment levels. This divergence is especially evident between months 7 and 13, offering insurance providers a valuable early warning window of over 6 months for proactive intervention. The resulting risk-based premiums range from \$201 (low risk) to \$5,942 (very high risk), reflecting behavioral risk profiles. The aggregate loss ratio of 71.5% supports profitable product operation and remains below the industry threshold of 80%. Early intervention efforts show that engaging high-risk customers (with over 40% predicted default probability) could prevent approximately 182 defaults (representing 30% of the current 607 defaults) and yield a 650% return on intervention investment through avoided losses claims.

3. Data and Methodology

3.1 Data Source and Description:

This analysis uses data from the American Express Default Prediction competition on Kaggle (American Express, 2022). The dataset includes credit card account details for about 460,000 US customers over a 13-month period from March 2017 to March 2018. To optimize computation and facilitate detailed analysis within project timelines, we

sampled a random subset of 10,000 customers, maintaining the original default rate distribution. This sampling ensures representativeness and supports iterative model development and testing.

The dataset features 190 anonymized variables categorized into five groups, identified by prefix codes: D_* (delinquency variables), S_* (spending behavior), P_* (payment variables), B_* (balance data), and R_* (risk indicators). Each customer has multiple monthly account snapshots, creating a panel data structure suitable for time-series feature engineering. The binary target indicates whether a customer defaulted (defined as 120+ days past due) within 18 months after the observation period.

Class imbalance exists in the dataset, with about 26% of customers labeled as defaulters, reflecting typical credit card default rates in the US. All features are fully anonymized, with original variable names and scales replaced by generic labels. The level of aggregation provides enough detail for behavioral analysis while avoiding the computational load of transaction-level data.

3.2 Feature Engineering: Time-Series Behavioral Indicators:

Traditional credit risk models mainly depend on cross-sectional features measured at a single point in time. To better understand behavioral changes and spot emerging financial distress patterns, time-series derived features that capture trends, volatility, and trajectories over the 13-month observation period were developed. This method converts raw panel data into customer-level features that reflect behavioral dynamics instead of just averaging past data.

For payment variables (P_* features), percentage changes were calculated from one period to the next to detect speeding up or slowing down payment behaviors. The absolute differences between consecutive months were also captured to catch sudden shifts, and 3-month rolling averages were used to smooth out short-term noise while keeping medium-term trends. These transformations help the model distinguish customers with steadily declining payments from those with temporary changes due to seasonal spending or irregular income.

Balance variables (B_* features) were treated similarly, focusing on credit utilization growth rates and balance volatility. Three-month rolling standard deviations measure

payment consistency, as irregular payment amounts often indicate financial stress even if average payments are stable. For spending variables (S_* features), we calculated growth rates and trend slopes to identify whether spending is accelerating beyond income, a common pattern among future defaulters.

After creating time-series features at the statement level, we aggregated them to the customer level using multiple statistics. For each feature, we computed mean, standard deviation, minimum, and maximum across the observation period. This approach captures both central tendency and dispersion, allowing models to differentiate customers with consistently poor payment habits from those showing worsening trends despite acceptable historical averages.

3.3 Modeling Approaches:

We evaluated three machine learning approaches representing different modeling philosophies: logistic regression with time-series features (linear baseline), random forest (ensemble tree-based), and gradient boosting (sequential ensemble). All models were trained on identical feature sets using 70-30 stratified train-test splits to ensure consistent evaluation. Logistic regression provides an interpretable baseline and enables assessment of individual feature contributions through coefficient inspection. We applied L2 regularization and balanced class weights to address class imbalance. Random forest offers robustness to feature interactions and nonlinear relationships through ensemble averaging across 100 decision trees with a maximum depth of 10 to prevent overfitting. Gradient boosting sequentially builds trees to correct previous errors, often achieving superior performance on tabular data. We implemented 100 boosting rounds with a maximum depth of 5 and a learning rate of 0.1. Model evaluation prioritized Area Under the ROC Curve (AUC) as the primary metric, as it captures discrimination ability across all classification thresholds which is critical for insurance applications where intervention thresholds may vary based on program resources and risk tolerance.

3.4 Insurance Framework: Premium Calculation and Intervention Economics

To convert model predictions into practical insurance metrics, we created a straightforward actuarial framework based on standard industry practices for consumer credit insurance. Expected loss per customer equals predicted default probability multiplied by an average claim amount of \$5,000, representing a conservative estimate below the current US average credit card balance of approximately \$6,500-\$7,300 (LendingTree, 2025; TransUnion, 2025). Premiums are set at 1.40 times expected loss to account for profit margins and administrative expenses, consistent with insurance industry standards where combined ratios (loss ratio plus expense ratio) typically target values below 100% to ensure profitable operations (Insurance Training Center, 2023).

Customers are divided into four risk tiers (Low, Medium, High, Very High) using predicted probability quartiles, enabling risk-based pricing that reflects actual actuarial exposure. We estimated intervention economics using reasonable industry assumptions: \$200 cost per outreach attempt (including staff time, communication expenses, and program administration), 30% success rate in preventing defaults among contacted high-risk customers, and \$5,000 avoided claim cost per prevented default. These assumptions reflect conservative estimates based on financial hardship program effectiveness reported in consumer finance literature, though actual intervention success rates may vary based on program design and customer engagement levels. Return on investment quantifies net financial benefit relative to intervention program costs.

4. RESULTS

4.1 Model Performance and Comparison:

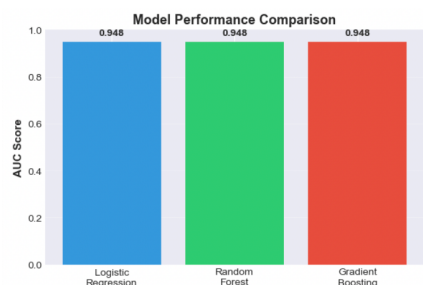
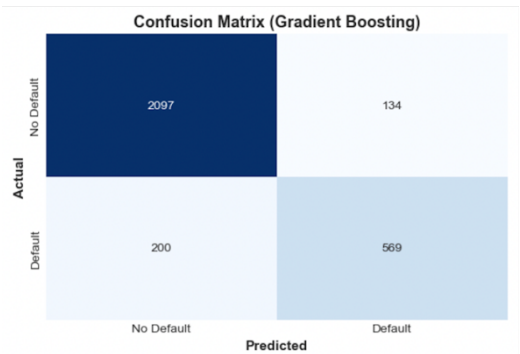
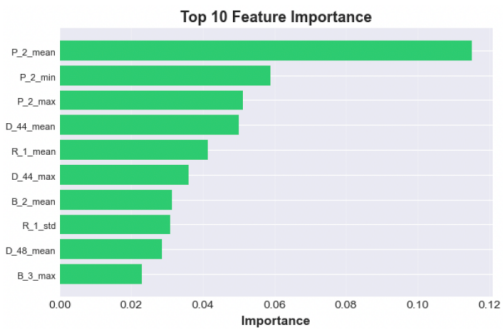


Figure on the left displays a comparison of three machine learning methods for default prediction. All three models show very similar performance, with Logistic Regression, Random Forest, and Gradient Boosting each reaching an AUC of 0.948 (94.8%). This

consistent performance across different models indicates that the behavioral features derived from time-series data capture reliable and consistent signals that various algorithms can utilize effectively. The strong results across all models support the idea that behavioral trajectories provide significant predictive information beyond static, point-in-time data.

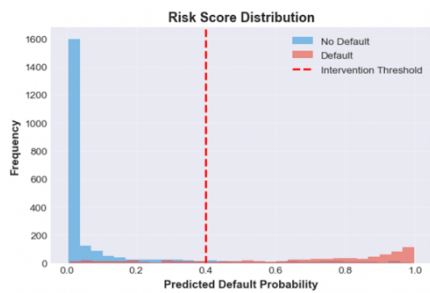
Feature-importance analysis of the Random Forest model indicates that payment-related variables are the most influential predictors. The mean of payment variable P_2 is the top feature, followed by its minimum and maximum values. Indicators of delinquency (D_44, D_48) and account balances (B_2, B_3) also rank highly, highlighting that payment regularity, balance management, and past delinquency are key risk signals. The importance of both average values and measures of variability (minimums, maximums, standard deviations) underlines the value of our feature engineering approach, which captures behavioral variability as well as average behavior.



The confusion matrix for Gradient Boosting indicates robust classification results, with 2,097 true negatives and 569 true positives correctly identified. The model produces 134 false positives (non-defaulters wrongly marked as high risk) and 200 false negatives (defaulters missed).

From an insurance standpoint, false negatives mean missed opportunities for intervention, potentially leading to unanticipated claims, while false positives could result in overly conservative pricing or unnecessary resource deployment. The 200 false negatives (roughly 26% of actual defaulters) highlight areas for improvement, but overall accuracy remains strong for commercial insurance purposes.

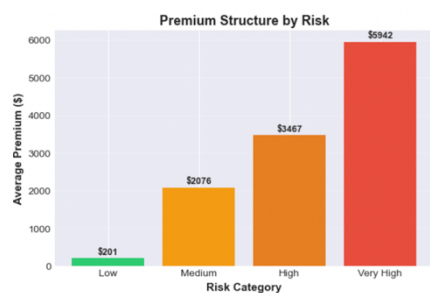
4.2 Risk-Based Premium Structure and Customer Segmentation:



The risk score distribution shows how well the model separates defaulters from non-defaulters. The bimodal pattern indicates most non-defaulters score low on predicted risk (0.0-0.2), while defaulters have higher and more varied risk scores. The intervention threshold set at 0.4 (40% predicted default probability), marked

by a red dashed line, identifies the point above which proactive customer engagement is economically justified based on cost-benefit analysis. About 11.4% of the portfolio exceeds this threshold comprising of 8.0% high-risk and 3.4% very high-risk customers forms the target group for early intervention programs.

Risk-based premium structures show considerable variation across risk categories.



Low-risk customers, making up 67.0% of the portfolio, would pay approximately \$201 annually, reflecting their low expected losses and keeping coverage affordable for most borrowers. Medium-risk customers (21.6%) face premiums of \$2,076, a tenfold increase from low-risk pricing. High-risk (8.0%) customers are charged

\$3,467, while very high-risk (3.4%) customers face premiums of \$5,942. This structure aligns premiums with expected losses, ensuring actuarial soundness, but may also price very high-risk customers out of voluntary coverage posing a challenge that insurers address through product design choices like risk mitigation requirements or mandatory coverage.

The customer distribution pie chart indicates that two-thirds of the portfolio falls into the low-risk group, supporting sustainable insurance operations. This concentration allows for cross-subsidization, helping keep average premiums affordable while maintaining profitability.

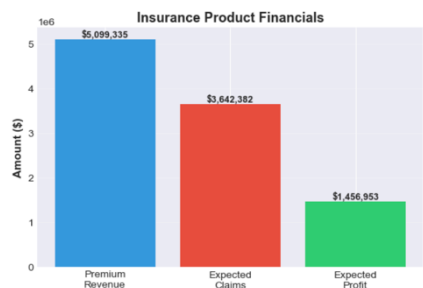
Nonetheless, 11.4% of customers in high- and



very-high-risk groups pose disproportionate claim risks, requiring either sufficiently high premiums or active risk mitigation through intervention programs.

4.3 Insurance Product Financial Viability:

The financial analysis confirms that the proposed insurance product structure is

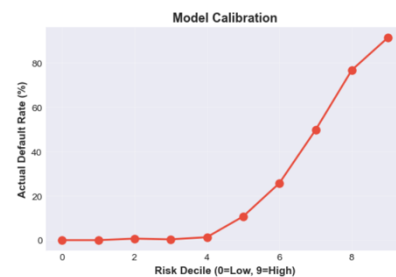


financially sustainable. Total premium revenue of \$5,099,335 from the 10,000 customers significantly surpasses expected claims of \$3,642,382, resulting in an expected profit of \$1,456,953. The resulting loss ratio of 71.5% (claims divided by premiums) is well within industry norms, as credit insurance products

generally aim for loss ratios between 70-80% to balance competitiveness and profitability.

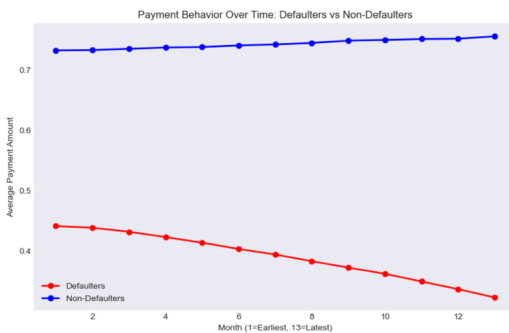
This 28.5% profit margin (\$1,456,953 profit on \$5,099,335 premiums) covers not only insurer profits but also administrative costs, claims processing, regulatory compliance, and capital requirements. The demonstrated financial viability suggests that

insurance products based on behavioral monitoring can be commercially sustainable while providing meaningful coverage to borrowers. The sizable profit margin also offers insurers the flexibility to invest in intervention programs, improve customer service, or lower premiums to boost market competitiveness.



4.4 Behavioral Pattern Analysis: Payment Trajectories Over Time:

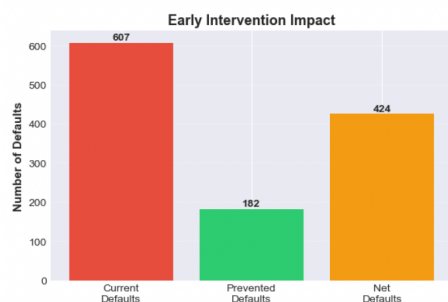
This figure illustrates the changes in payment behavior over 13 months for both



Significant divergence detected at Month 1
This is approximately 12 months before the end of observation

defaulters and non-defaulters. The data shows clear differences: defaulters' average payments decrease steadily from 0.44 in Month 1 to 0.32 in Month 13, a 27% drop. This gradual decline implies that financial difficulties develop over time, enabling early detection and intervention.

4.5 Early Intervention Impact Analysis:



The intervention impact analysis measures the possible benefits of proactive customer engagement based on behavioral risk scores. Out of a sample of 10,000 customers, the model identifies 607 who ultimately defaulted during the observation period.

Targeting high-risk customers (those with a probability above 0.4) with financial counseling, adjusted payment plans, or hardship programs could potentially prevent 182 defaults, assuming a conservative 30% intervention success rate. While specific success rates vary based on program design and customer engagement (Bankrate, 2025), this conservative estimate provides a baseline for evaluating the economic viability of early intervention programs.

If each contacted customer costs \$200 to reach and the average claim savings per prevented default is \$5,000, the program would likely deliver a strong return on investment. Preventing 182 defaults would save \$910,000 in claims ($\$5,000 \times 182$), while contacting about 700 high-risk customers would cost roughly \$140,000 ($\200×700). This results in net savings of \$770,000 and an ROI of over 650% (\$770,000 profit divided by \$140,000 expense).

These financial figures prove that behavioral monitoring techniques offer value beyond better prediction, they support cost-effective intervention efforts that reduce losses and

benefit customers. The financial argument for early action is even more compelling when considering that preventing defaults also helps retain customer relationships, maintain credit bureau scores, and avoid potential negative publicity from aggressive collection methods.

5. Recommendations

This analysis offers several practical recommendations for designing payment protection insurance (PPI) products. These include implementing behavioral monitoring systems, developing proactive intervention programs, and adopting dynamic pricing structures that reflect actual risk levels rather than relying on static assessments at policy start.

Firstly, insurers should adopt continuous behavioral monitoring that tracks customer payment patterns throughout the policy's life. Unlike traditional PPI products that assess risk only at purchase, this study shows that payment behavior can change significantly over time. Early signs of financial stress, such as declining payments, can be detected through ongoing monitoring, allowing insurers to identify at-risk customers before missed payments occur. Automated systems can analyze payment trajectories using existing data from credit card providers, which already have access to relevant payment information.

Secondly, insurers should introduce structured early intervention programs based on behavioral risk scores. The analysis indicates that engaging high-risk customers proactively could prevent about 30% of defaults and yield a 650% return on investment. Interventions could include financial education for those showing early warning signs, and more intensive measures such as modified payment plans or access to financial counseling for customers with more severe risks. The goal is to intervene early, before a default happens, rather than reacting afterward.

Thirdly, risk-based premium pricing should replace flat-rate or minimally differentiated prices. The study shows that predicted default probabilities vary widely, with premiums ranging from \$201 for low-risk individuals to \$5,942 for very high-risk ones, based on behavioral patterns. Transitioning to true risk-based pricing ensures premiums are

aligned with actual risk, making coverage more affordable for low-risk customers and financially sustainable for insurers handling higher-risk groups.

However, very high premiums for at-risk customers can raise fairness issues, especially if costs become unaffordable. A solution is to design hybrid products that combine insurance with participation in financial wellness programs. For example, at-risk customers could benefit from reduced premiums in exchange for completing financial counseling or maintaining emergency savings, thereby actively reducing their risk through behavioral change.

Finally, insurers should consider implementing dynamic premium adjustments based on ongoing behavioral data. Instead of fixed premiums at policy start, periodic reviews, quarterly or semi-annual, could allow adjustments. Customers improving their payment behavior might receive discounts, while those deteriorating could face premium increases or be enrolled in targeted intervention programs. This approach incentivizes healthy financial habits while ensuring premiums accurately reflect current risk levels.

6. Limitations and Future Improvements

Several limitations of this analysis should be acknowledged. First, the study relies on a sample of 10,000 customers from a larger dataset of 460,000. While the sampling preserved the overall default rate distribution, it may not fully capture regional, demographic, or product-specific variations present in the full dataset. Future research should validate these findings with the complete data and explore behavioral differences across customer segments based on geography, age, income, or credit history.

Second, the insurance framework includes simplifying assumptions. The average claim of \$5,000 offers a rough estimate of typical credit card balances at default, but actual claims vary widely by credit limits, utilization, and PPI policy terms. The assumed intervention success rate of 30% is an assumption based on general industry data rather than specific experimental evidence. Future studies should incorporate detailed actuarial models with product-specific claim data and conduct controlled trials to measure real intervention effectiveness.

Third, this analysis uses data from 2017-2018 in the United States. Economic conditions, regulations, and consumer behaviors differ across countries and periods.

The observed behavioral patterns may not apply elsewhere or post-pandemic. Future research should assess whether the predictive power of behavioral time-series features remains stable across various economic cycles, regions, and regulatory environments. Fourth, the modeling approach does not explicitly employ causal inference methods to differentiate correlation from causation. While declining payments predict default, it's unclear whether declines cause defaults or both result from financial stress. Future work could use techniques like instrumental variables to identify causal links between behavioral changes and default risk.

Finally, this study omits practical implementation challenges such as data infrastructure, regulatory approval, and customer communication. Deploying behavioral monitoring requires advanced technology, governance, and compliance, which vary by jurisdiction. Future research should evaluate the operational feasibility of large-scale deployment and customer response to dynamic pricing and intervention programs through pilot tests and field experiments.

7. Conclusion

This study shows that payment protection insurance can be greatly enhanced by continuously monitoring credit card payment and spending behaviors. Analyzing data from around 10,000 American Express customers over 13 months, machine learning models were built to predict default risk with 94.8% AUC by using time-series behavioral features. The key discovery is that defaulters follow very different payment patterns from non-defaulters, with payment amounts decreasing by 27% during the review period, while non-defaulters keep or increase their payments.

These behavioral insights open up practical opportunities for insurance providers. The analysis reveals that risk-based premium pricing, from \$201 to \$5,942 per customer, can be justified based on behavioral risk profiles, leading to a loss ratio of 71.5%, which supports financial sustainability and remains competitive industry-wide. Crucially, early intervention strategies aimed at high-risk customers could prevent about 30% of defaults and deliver a 650% return on investment through lower claim costs. This economic case proves that behavioral monitoring adds value not just by improving

prediction but also by enabling proactive risk management that benefits both insurers and customers.

The results advocate a fundamental change in how payment protection insurance should be designed and operated. Instead of seeing insurance as a passive risk transfer with static assessments at policy start, insurers should adopt active risk management practices that continuously monitor customer behavior and intervene early when signs of financial stress appear. Implementing this transformation demands technological investments in data integration and behavioral tracking, but the financial gains and customer benefits make the investment worthwhile.

More broadly, this research highlights that machine learning and behavioral analytics serve practical purposes beyond just improving prediction accuracy. Their real value lies in converting insights into operational changes such as dynamic pricing, intervention protocols, and customer engagement. As the payment protection insurance market is expected to grow to \$34.7 billion by 2033, insurers who develop effective behavioral monitoring systems will enjoy competitive advantages through better risk selection, fewer claims, and higher customer satisfaction compared to those relying solely on traditional assessments.

For customers under financial strain, behavioral monitoring integrated into insurance products offers additional benefits beyond claim payments after default. Early intervention triggered by emerging risk signals could help customers avoid default altogether through financial counseling, adjusted payment plans, or connections to hardship programs. This proactive strategy aligns insurer and customer interests by creating shared incentives to prevent financial decline rather than merely handling claims post-default. The result is a more sustainable insurance model that reduces both insurer losses and customer financial hardship through timely support and intervention.

Citations:

- Abikoye, B., Agorbia-Atta, C., Noriega, J. P., Rivera, L. A., & Herrera, J. A. (2024). How artificial intelligence and machine learning are transforming credit risk prediction in the financial sector. *International Journal of Science and Research Archive*, 12(2), 1467-1476. <https://doi.org/10.30574/ijrsra.2024.12.2.1467>
- American Express. (2022). American Express - Default Prediction [Data set]. Kaggle. <https://kaggle.com/competitions/amex-default-prediction>
- Bankrate. (2025). What is a credit card hardship program? <https://www.bankrate.com/credit-cards/advice/what-is-a-credit-card-hardship-program/>
- DataIntel. (2024). Payment protection insurance market research report 2033. <https://dataintel.com/report/payment-protection-insurance-market>
- Heterogeneity in demand for credit payment protection insurance (Dissertation). Retrieved from <https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-214454>
- Insurance Training Center. (2023). Understanding combined ratio. <https://insurancetrainingcenter.com/resource/understanding-combined-ratio/>
- LendingTree. (2025). 2025 credit card debt statistics. <https://www.lendingtree.com/credit-cards/study/credit-card-debt-statistics/>
- Meriläinen, E. (2023). Heterogeneity in demand for credit payment protection insurance [Bachelor's thesis, Umeå University]. DiVA. <https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-214454>
- Nayak, S. (2024). Developing predictive models for financial stability: Integrating behavioral analytics into credit risk management. *Journal of Artificial Intelligence & Cloud Computing*, 3, E173. [https://doi.org/10.47363/JAICC/2024\(3\)E173](https://doi.org/10.47363/JAICC/2024(3)E173)
- TransUnion. (2025). U.S. credit industry insights report, Q3 2025. Retrieved from <https://www.transunion.com>